AI Security

Optimization for Black-box Attacks

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POSTECH

April 14, 2026

Outline

1. Introduction
2. Prompt Tuning
3. Zeroth Order Optimization
4. Policy Optimization

Need for Black-box Attacks

Main Question

How can we find an adversarial prompt in this black-box setup?

- Closed models (*e.g.*, GPT-5) is the most interesting attack targets
- However, we cannot have access to their model weights.
- But, still we can obtain responses from them (*i.e.*, only allows forward-pass).

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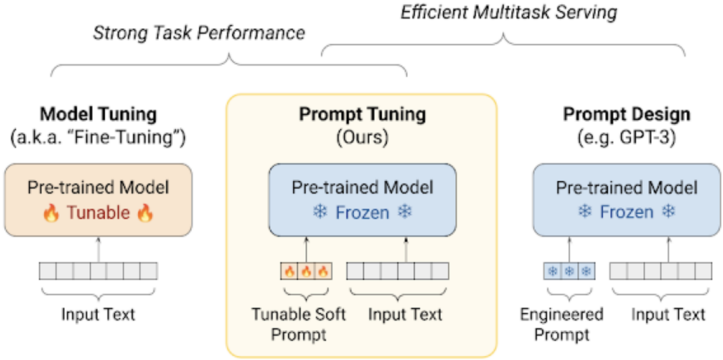
Why Prompt Tuning?



- Prompt tuning is a key to guide LLMs to have desired responses.
- We often build hard prompts for it.
- Prompt Tuning [Lester et al., 2021] finds soft input embedding vectors, instead of hard prompts.
- This is used for maximizing performance but can be applicable for attacks.

Prompt Tuning [Lester et al., 2021]

Objective

$$\min_{\mathbf{E}} -\sum_{n=1}^{N} \log p_\theta(\mathbf{y}_n \mid \mathbf{E}, \mathbf{x}_n)$$

- D: the embedding vector dimension
- K: the number of tuning tokens
- $\mathbf{E} \in \mathbb{R}^{D \times K}$: a set of input embedding vectors
- p_θ: an LLM
- In attacks where $\mathbf{y}_n$ is a harmful response, is it reasonable for attackers to manipulate $\mathbf{E}$?

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What and Why Zeroth Order Optimization?

Question

Can we optimize parameters without gradients (or with only forward passes)?

- zeroth order optimization (ZO) methods optimizes a model by estimating gradients using forward passes.
- Why ZO?
  - no gradients – memory efficient
  - non-differentiable criteria, *e.g.*, rule-based safety standards or F1 score
- Why ZO in attacks?
  - may be useful for black-box models

Simultaneous Perturbation Stochastic Approximation (SPSA)

Definition (SPSA)

SPSA estimate the gradient of a loss function $\mathcal{L}$ for a model θ on a minibatch $\mathcal{B}$ as follows:

$$\hat{\nabla}\mathcal{L}(\theta;\mathcal{B}) = \frac{\mathcal{L}(\theta+\varepsilon z;\mathcal{B}) - \mathcal{L}(\theta-\varepsilon z;\mathcal{B})}{2\varepsilon} z \approx zz^T \nabla\mathcal{L}(\theta;\mathcal{B}),$$

where $z \sim \mathcal{N}(0, I_d)$ and ε is the perturbation scale.

- Intuitively, this estimates a gradient (*e.g.*, imagine then $d = 1$)
- Here, you can derive $\approx$ via the first order Taylor expansion.
- How many forward passes do we need?
- How much memory do we need?
- Can you imagine the convergence speed of SPSA-based SGD?
- Is this unbiased estimator?

MeZO [Malladi et al., 2023]

Algorithm 1: MeZO

Require: parameters $\boldsymbol{\theta} \in \mathbb{R}^d$, loss $\mathcal{L} : \mathbb{R}^d \to \mathbb{R}$, step budget T, perturbation scale ϵ, batch size B, learning rate schedule $\{\eta_t\}$

for $t = 1, ..., T$ **do**
  Sample batch $\mathcal{B} \subset \mathcal{D}$ and random seed s
  $\boldsymbol{\theta} \leftarrow$ `PerturbParameters(`$\boldsymbol{\theta}, \epsilon, s$`)`
  $\ell_+ \leftarrow \mathcal{L}(\boldsymbol{\theta}; \mathcal{B})$
  $\boldsymbol{\theta} \leftarrow$ `PerturbParameters(`$\boldsymbol{\theta}, -2\epsilon, s$`)`
  $\ell_- \leftarrow \mathcal{L}(\boldsymbol{\theta}; \mathcal{B})$
  $\boldsymbol{\theta} \leftarrow$ `PerturbParameters(`$\boldsymbol{\theta}, \epsilon, s$`)` ▷ `Reset parameters before descent`
  `projected_grad` $\leftarrow (\ell_+ - \ell_-)/(2\epsilon)$
  Reset random number generator with seed s ▷ `For sampling` z
  for $\theta_i \in \boldsymbol{\theta}$ **do**
    $z \sim \mathcal{N}(0, 1)$
    $\theta_i \leftarrow \theta_i - \eta_t *$ `projected_grad` $* z$
  end
end

Subroutine `PerturbParameters(`$\boldsymbol{\theta}, \epsilon, s$`)`
  Reset random number generator with seed s ▷ `For sampling` z
  for $\theta_i \in \boldsymbol{\theta}$ **do**
    $z \sim \mathcal{N}(0, 1)$
    $\theta_i \leftarrow \theta_i + \epsilon z$ ▷ `Modify parameters in place`
  end
  return $\boldsymbol{\theta}$

- MeZO: Memory-efficient Zeroth order Optimizer
- Need only $\mathcal{O}(\mathbb{R}^d)$ memory

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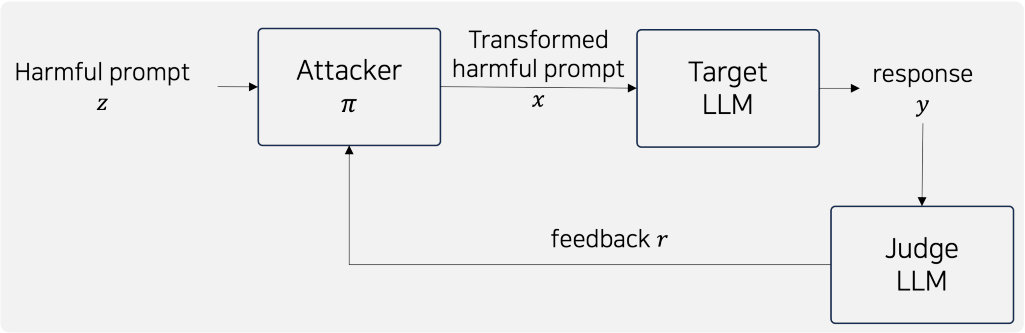
Why Policy Optimization?

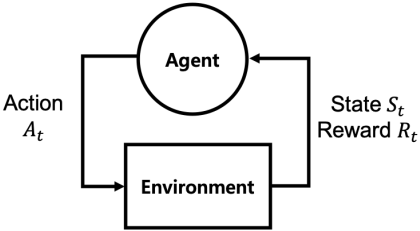


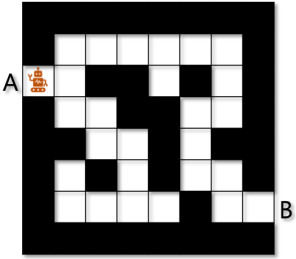
- We often see only the victim model's responses – black box setups
- Finding a good attacker = learning an attacker via RL – very natural

Learning from Interaction

Learning protocol:

- The agent observes an initial state S_0
- For each step $t = 0, 1, 2, \ldots$
- The agent executes an action A_t given the observed state S_t
- The agent observes reward R_{t+1} and the next state S_{t+1}





Markov Decision Process (MDP)

Definition (informal)

A Markov decision process (MDP) is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, p, d_0)$, where

- $\mathcal{S}$: a set of states
- $\mathcal{A}$: a set of actions (or $\mathcal{A}(s)$: a set of actions given a state s)
- $\mathcal{R}$: a set of reward values
- $p : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S} \times \mathcal{R})$: dynamics or a kernel
- d_0: an initial state distribution

For example,

- $s_0 \sim d_0$: an initial location of a robot
- $a_0 \in \mathcal{A}$: "move to the right"
- $p(s_1, r_1 \mid s_0, a_0)$: a proability distribution over next states and rewards

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- In MDP, p completely characterizes the environment's dynamcis due to the Markov property.
  - See the next slide.

Markov Property

Definition

The dyanmics $p : (\mathcal{S} \times \mathcal{A})^* \to \Delta(\mathcal{S} \times \mathcal{R})$ has a Markov property if

$$\overbrace{\mathbb{P}\{S_{t+1} = s_{t+1}, R_{t+1} = r_{t+1} \,|\, S_{1:t} = s_{1:t}, A_{1:t} = a_{1:t}\}}^{p(s_{t+1}, r_{t+1} | s_{1:t}, a_{1:t})} = \underbrace{\mathbb{P}\{S_{t+1} = s_{t+1}, R_{t+1} = r_{t+1} \mid S_t = s_t, A_t = a_t\}}_{p(s_{t+1}, r_{t+1} | s_t, a_t)}.$$

- This means S_t include perfect information on previous action-environment interaction.
- Here, we assume that the dynamcis does not change over time t (=homogeneous Markov chain).

RL Goal

Goal – informal

The goal of RL is to find an optimal "policy" that maximizes an "expected cumulative reward" (or an "expected return").

Note that

- the goal is NOT maximizing immediate rewards.
- the expected return $\approx$ an estimate of rewards that the agent will recieve in future.

We will see the answers of the following questions:

- What is the "policy"?
- What is the "expected return"?

Policy

Definition

A policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ is a mapping from states to probabilities of selecting each possible action.

- The policy can be deterministic – in this case we simply denote it by $\pi : \mathcal{S} \to \mathcal{A}$.
- The policy is modeled via a neural net – this is our target to learn

Example:

	s_1	s_2	s_3
$\pi_1(a_i \| s_i)$	a_1 a_2 a_3 1.0 0 0	a_1 a_2 a_3 0.5 0.5 0	a_1 a_2 a_3 0.2 0.3 0.4
$\pi_2(a_i \| s_i)$	a_1 a_2 a_3 0 1.0 0	a_1 a_2 a_3 1/3 1/3 1/3	a_1 a_2 a_3 0 0.5 0.5

Trajectory

Trajectory

A trajectory τ is a sequence of states, actions, and rewards that is obtained by the interaction of a policy π with its environment, *i.e.*,

$$\tau := (S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, \ldots).$$

- Often called a roll-out
- Is the trajectory a random variable? If so, where are sources of randomness?

Return

Definition

We say that G_t is a return with a finite horizon T, defined as

$$G_t := R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots + \gamma^{T-t-1} R_T$$

where R_k is a reward at time k and γ is a discount factor.

- R_k: the immediate reward
- The discount factor quantifies the appreciation on future rewards.
  - $\gamma = 0$: only consider the immediate reward but ignore all future rewards.
  - $0 \leq \gamma \leq 1$: consider the immediate reward and discount the appreciation more on more "distant" future rewards.
  - $\gamma = 1$: equallty appreciate all future rewards, including the immediate reward.
- The return has a recursive structure, *i.e.*,

$$G_t = R_{t+1} + \gamma G_{t+1}.$$

RL Goal

Goal – informal

The goal of RL is to find an optimal policy that maximizes an expected return, *i.e.*,

$$\max_{\pi} \mathbb{E} G_0.$$

- The expectation is taken over trajectories.
- How to find the optimal policy?
  - Policy optimization
  - "Q" function and "V" function

Action-Value Function (="Q" Function)

State-Value Function (="V" Function)

Policy Optimization – A Bird's-eye View

Reference I

B. Lester, R. Al-Rfou, and N. Constant. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the 2021 conference on empirical methods in natural language processing*, pages 3045–3059, 2021.

S. Malladi, T. Gao, E. Nichani, A. Damian, J. D. Lee, D. Chen, and S. Arora. Fine-tuning language models with just forward passes. *Advances in Neural Information Processing Systems*, 36:53038–53075, 2023.